

CS215 Assignment-1

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1. Normal Sampling

Task A:

Solution: Consider the random variable $Y := F_X(X)$ in $[0, 1]$.

To find its distribution, we compute the CDF of Y , denoted by $F_Y(y)$. By definition:

$$F_Y(y) = P(Y \leq y) = P(F_X(X) \leq y).$$

ATQ, F_X is invertible, ie, there exists a function F_X^{-1} which is inverse of F_X .

Since $Y = F_X(X)$, the event $F_X(X) \leq y$ is equivalent to the event $X \leq F_X^{-1}(y)$. Therefore, we have:

$$F_Y(y) = P(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) \quad (\text{by definition of CDF for } X)$$

$$\forall y \in [0, 1] \quad F_X(F_X^{-1}(y)) = y \implies F_Y(y) = y.$$

The CDF of Y is $F_Y(y) = y$ for $y \in [0, 1]$, which is exactly the CDF of a uniformly distributed random variable on $[0, 1]$.

Thus, Y is uniformly distributed on $[0, 1]$.

Task B:

Solution:

Assumption: We know the CDF of X .

Given a random variable Y uniformly distributed over $[0, 1]$, an algorithm $\mathcal{A}$ that transforms Y into a sample from a random variable with the CDF of X can be constructed as follows:

Let $Y \sim \text{Uniform}(0, 1)$. Suppose F_X is the CDF of the desired random variable X , which is continuous and invertible.

Algorithm $\mathcal{A}$:

Transform the uniform random variable Y by applying the inverse of the CDF of X .

$$\mathcal{A}(y) = F_X^{-1}(y).$$

Proof of Correctness:

Consider the CDF of $Z = F_X^{-1}(Y)$.

$$F_Z(u) = P(Z \leq u) = P(F_X^{-1}(Y) \leq u).$$

Since F_X is continuous and invertible, this is equivalent to:

$$P(F_X^{-1}(Y) \leq u) = P(Y \leq F_X(u)).$$

Because Y is uniformly distributed over $[0, 1]$, we have:

$$P(Y \leq F_X(u)) = F_X(u).$$

Thus, the CDF of Z is $F_Z(u) = F_X(u)$

Hence, Z (the output of the algorithm $\mathcal{A}$) has the same CDF as X .